



College of Engineering, Construction and Living Sciences
Bachelor of Information Technology
ID721001: Mobile Application Development
Level 7, Credits 15
Project

Assessment Overview

In this **individual** assessment, you will develop two mobile applications using **React Native** and **Expo**.

Learning Outcomes

At the successful completion of this course, learners will be able to:

1. Implement and publish complete, non-trivial, industry-standard mobile applications following sound architectural and code-quality standards.
2. Identify relevant use cases for a mobile computing scenario and incorporate them into an effective user experience design.
3. Follow industry standard software engineering practice in the design of mobile applications.

Assessments

Assessment	Weighting	Due Date	Learning Outcomes
Practical	20%	Sunday 9th November at 11.59 PM	1, 2
Project	80%	Sunday 9th November at 11.59 PM	1, 2, 3

Conditions of Assessment

You will complete this assessment mostly during your learner-managed time. However, there will be time during class to discuss the requirements and your progress on this assessment. This assessment will need to be completed by **Sunday 9th November at 11.59 PM**.

Pass Criteria

This assessment is criterion-referenced (CRA) with a cumulative pass mark of **50%** over all assessments in **ID721001: Mobile Application Development**.

Submission

You **must** submit all files via **GitHub Classroom**. Here is the URL to the repository you will use for your submission – https://classroom.github.com/a/Mq0MeQ_x. If you do not have not one, create a **.gitignore** and add the ignored files in this resource - <https://raw.githubusercontent.com/github/gitignore/main/Node.gitignore>.

Create a branch called **project**. The latest files in the **project** branch will be used to mark against the marking rubric. Please test before you submit. Partial marks **will not** be given for incomplete functionality. Late submissions will incur a **10% penalty per day**, rolling over at **12.00 AM**.

Authenticity

All parts of your submitted assessment **must** be completely your work. The use of **AI tools** is not permitted in this assessment. You need to demonstrate to the course lecturer that you can meet the learning outcomes for this assessment. You **must** complete the <https://moodle.op.ac.nz/mod/page/view.php?id=1178172> form and put this in the root of your repository. Failure to do this may result in a mark of **zero** for this assessment.

Policy on Submissions, Extensions, Resubmissions and Resits

The school's process concerning submissions, extensions, resubmissions and resits complies with **Otago Polytechnic** policies. Learners can view policies on the **Otago Polytechnic** website located at <https://www.op.ac.nz/about-us/governance-and-management/policies>.

Extensions

Familiarise yourself with the assessment due date. Extensions will **only** be granted if you are unable to complete the assessment by the due date because of **unforeseen circumstances outside your control**. The length of the extension granted will depend on the circumstances and **must** be negotiated with the course lecturer before the assessment due date. A medical certificate or support letter may be needed. Extensions will not be granted on the due date and for poor time management or pressure of other assessments.

Resits

Resits and reassessments are not applicable in **ID721001: Mobile Application Development**.

Instructions

Note: If you wish to develop different applications than those specified below, you **must** get approval from the course lecturer first.

Functionality - Learning Outcomes 1, 2 (20%)

- Application v1 with the following functionality:
 - Data is fetched from a local **JSON** file. You have been provided with a sample file called **application-development-data.json**.
 - App icon and splash screen.
 - Styled using **Tailwind CSS**.
 - **Recipe Details Screen Features:**
 - * Display the recipe's name, cuisine, image, ingredients, difficulty, prep time, cook time, servings, tags, nutrition, average rating and reviews.
 - * Display a heart icon in the top-right corner. Tapping it adds or removes the recipe and saves the changes.
 - * Display a plus icon next to the heart icon. Tapping it adds or removes the recipe's ingredients and saves the changes, ensuring no duplicate ingredients are stored.
 - **Daily Specials Screen Features:**
 - * Display a list of five randomly selected recipes with difficulty level Easy or Medium.
 - * Each item should show the recipe's name, cuisine, image and average rating.
 - * Tapping an item should navigate to the **Recipe Details Screen**.
 - **All Recipes Screen Features:**
 - * Display a list of all recipes.
 - * Each item should show the recipe's name, image, difficulty level, total time and average rating.
 - * Tapping an item should navigate to the **Recipe Details Screen**.
 - **Favourites Screen Features:**
 - * Display a list of saved favourite recipes.
 - * Each item should show the recipe's name, image and average rating.
 - * Tapping an item should navigate to the **Recipe Details Screen**.
 - **Ingredients Screen Features:**
 - * Display a list of saved ingredients grouped by category.
 - * Each item should display the ingredient name, quantity and category.
 - * Display a trash icon. Tapping it removes the ingredient and saves the changes.
- Application v2, which builds on v1 with the following functionality:
 - Select any four **Medium Features** from the list below:
 - * Filter recipes by cuisine, tags, difficulty level and ingredient categories.
 - * Search recipes by name, ingredient name or tag.
 - * Sort recipes by prep time, cook time, total time, difficulty level or average rating.
 - * Dark mode support with a theme toggle.
 - * Delete all saved favourite recipes or ingredients.
 - * Display recipes by times cooked or last cooked.
 - * Display a list of the last five recipes viewed.
 - Select any four **Hard Features** from the list below:
 - * Save and edit notes on recipes.

- * Add ratings and write reviews for recipes.
- * View combined ratings and user reviews.
- * Manage user profiles with preferences and dietary needs.
- * Add new recipes.
- * Update and delete existing recipes.
- * Meal planner to schedule recipes by day.
- Choose two features of your choice. These features should be different from the ones listed above.
- Create an **APK** for both v1 and v2.

Code Quality and Best Practices - Learning Outcomes 1, 3 (35%)

- Write clean, readable, and maintainable code.
- Use modular, reusable components and avoid code duplication.
- Keep code simple and easy to understand.
- Ensure the applications are responsive across different screen sizes and orientations.
- Optimise performance by managing resources efficiently.

Version Control - Learning Outcome 3 (5%)

- Regular commits are made throughout development, demonstrating meaningful progress.
- Commit messages are clear and descriptive.
- The repository shows a clean and logical commit history that reflects the development process of the applications.

Testing - Learning Outcome 3 (10%)

- Test the applications with three different users for each version (six users total), observing how they navigate and use features.
- Collect written feedback on usability issues, confusing UI and bugs.
- Write a brief report summarising the user testing, including:
 - Tester information (anonymised).
 - Tasks performed.
 - Positive feedback received.
 - Areas for improvement identified.
 - Bugs or errors found.
 - Suggestions on how feedback could be used to improve the applications (no requirement to implement).

Reflection - Learning Outcomes 1, 3 (5%)

- Write a short reflection (300-500 words) on your development process and learning throughout the project, including:
 - Challenges you faced and how you overcame them.
 - What you would do differently in future projects.
 - New skills, tools or practices you learned during the project.

Presentation - Learning Outcome 2 (5%)

- Record a 5-10 minute screen recording of your applications.
- Demonstrate the v1 and v2 features.
- Make sure the video is clear and easy to follow.
- Submit a link to the presentation in your repository's **README.md** file.